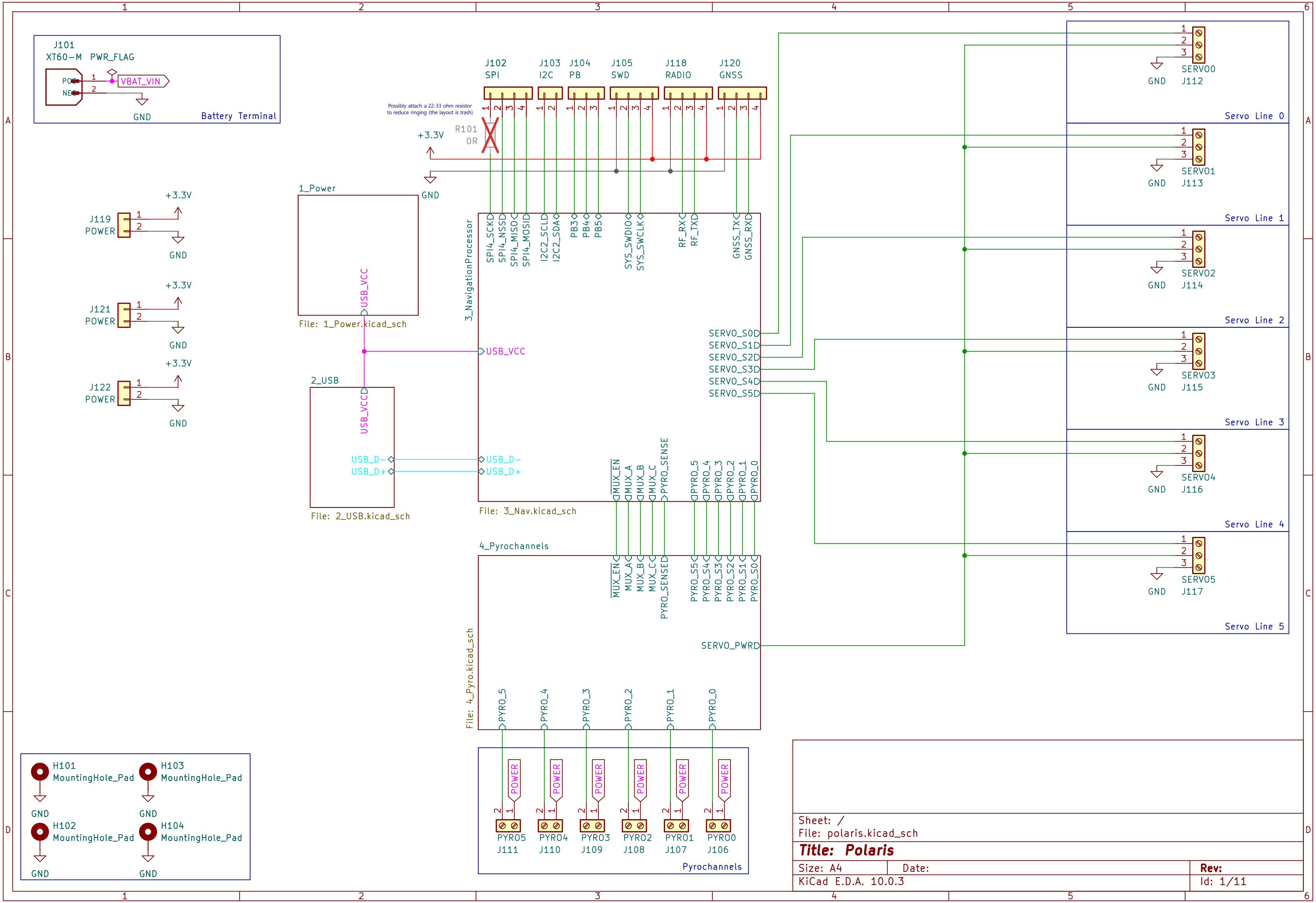

J101
XT60-M PWR_FLAG
VBAT_VIN
GND
Battery Terminal
J102
SPI
J103
I2C
J104
PB
J105
SWD
J118
RADIO
J120
GNSS
Possibly attach a 22-33 ohm resistor
to reduce ringing (the layout is trash)
R101
0R
+3.3V
GND
1_Power
USB_VCC
File: 1_Power.kicad_sch
2_USB
USB_VCC
USB_D-
USB_D+
File: 2_USB.kicad_sch
3_NavigationProcessor
SPI4_SCK
SPI4_NSS
SPI4_MISO
SPI4_MOSI
I2C2_SCL
I2C2_SDA
PB3
PB4
PB5
SYS_SWDIO
SYS_SWCLK
RF_RX
RF_TX
GNSS_TX
GNSS_RX
USB_VCC
USB_D-
USB_D+
SERVO_S0
SERVO_S1
SERVO_S2
SERVO_S3
SERVO_S4
SERVO_S5
MUX_EN
MUX_A
MUX_B
MUX_C
PYRO_SENSE
PYRO_5
PYRO_4
PYRO_3
PYRO_2
PYRO_1
PYRO_0
File: 3_Nav.kicad_sch
4_Pyrochannels
MUX_EN
MUX_A
MUX_B
MUX_C
PYRO_SENSE
PYRO_S5
PYRO_S4
PYRO_S3
PYRO_S2
PYRO_S1
PYRO_S0
SERVO_PWR
PYRO_5
PYRO_4
PYRO_3
PYRO_2
PYRO_1
PYRO_0
File: 4_Pyro.kicad_sch
J119
POWER
J121
POWER
J122
POWER
+3.3V
GND
SERVO0
J112
Servo Line 0
SERVO1
J113
Servo Line 1
SERVO2
J114
Servo Line 2
SERVO3
J115
Servo Line 3
SERVO4
J116
Servo Line 4
SERVO5
J117
Servo Line 5
POWER
PYRO5
J111
PYRO4
J110
PYRO3
J109
PYRO2
J108
PYRO1
J107
PYRO0
J106
Pyrochannels
H101
MountingHole_Pad
H103
MountingHole_Pad
H102
MountingHole_Pad
H104
MountingHole_Pad
GND
Sheet: /
File: polaris.kicad_sch
Title: Polaris
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.3
Id: 1/11

[1] Power

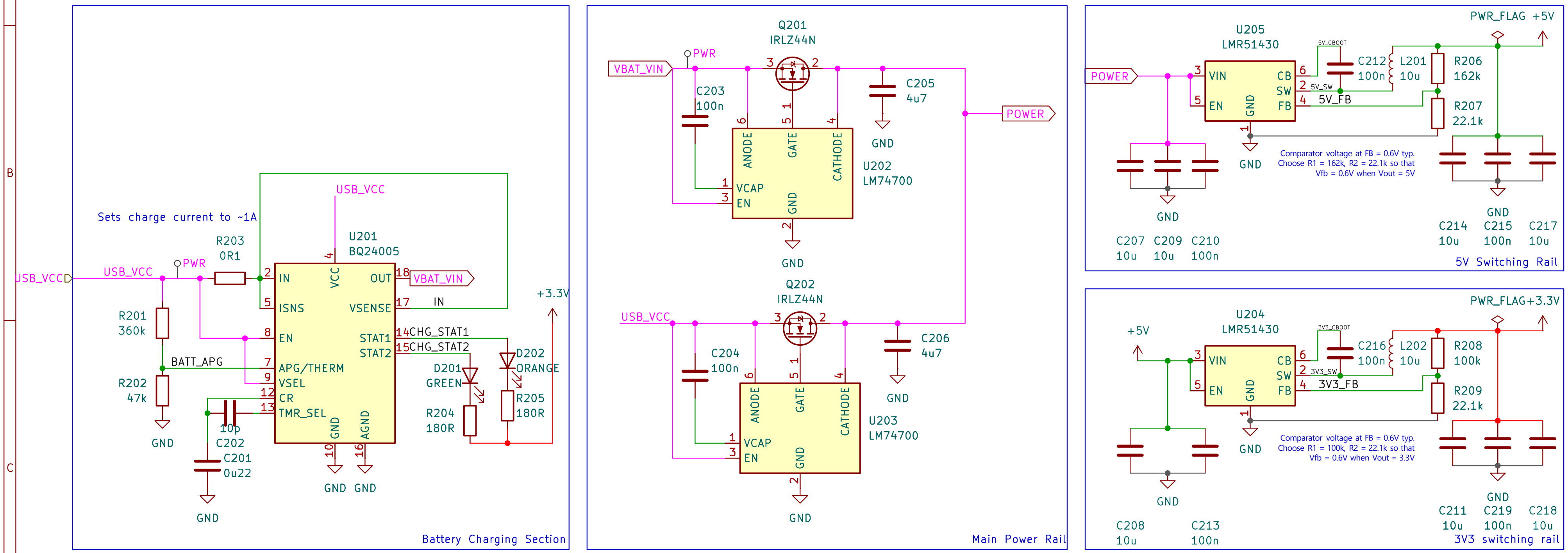
Output inductor calculations:
Lmin = (Vin_max - Vout) * Vout / Iout * K * Vin_max * f
Vin_max = 9V, Vout = 5V, Iout = 2A, K = 0.3, f = 500kHz
Lmin = ~7.4 uH. Choose nearest available 10 uH

Output capacitor calculations:
ΔVout_esr = K * Iout * ESR = 30 mV max
⇒ ESR = 0.05 Ω max

ΔVout_c = K * Iout / 8 * f * Cout = 30 mV max
⇒ Cout = 5 uF

Cout > 4 * (Ioh - Iol) / f * ΔVout_shoot
Using Ioh = 1.5A, Iol = 0.5A, ΔVout_shoot = 250 mV:
Cout > 32 uF
Using Cout > 32 uF at 50 mΩ ESR:
Choose Cout = 2 x 10 uF + 1 x 22 uF at 10V = 42 uF

Please remember to keep the buck converters far from the USB−C receptacle to reduce EMI



Output inductor calculations:
Lmin = (Vin_max - Vout) * Vout / Iout * K * Vin_max * f
Vin_max = 5.5V, Vout = 3.3V, Iout = 1.5A, K = 0.3, f = 500kHz
Lmin = ~5.9 uH. Choose nearest available 10 uH

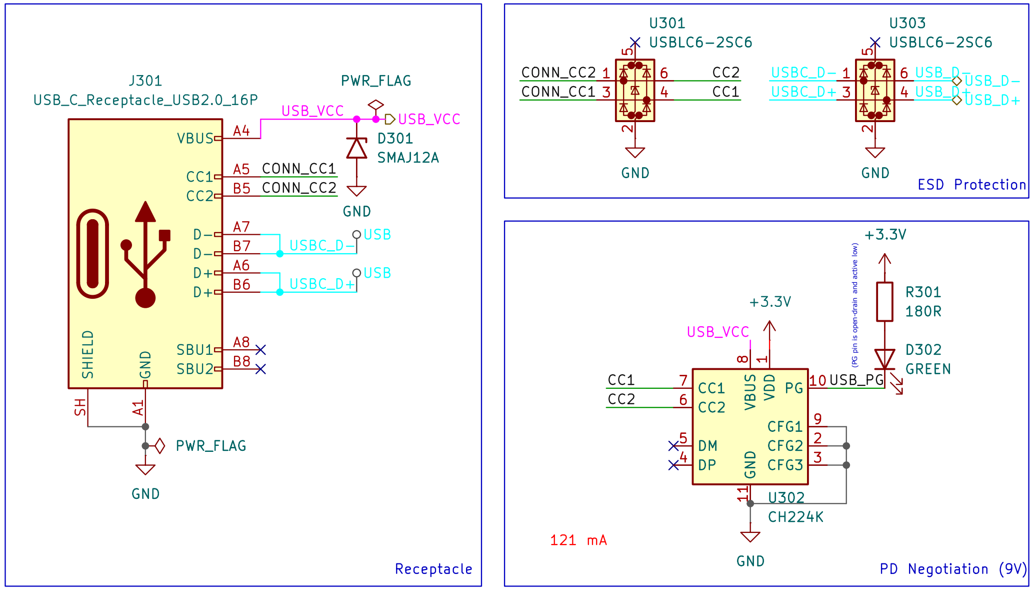
Output capacitor calculations:
ΔVout_esr = K * Iout * ESR = 30 mV max
⇒ ESR = 0.067 Ω max

ΔVout_c = K * Iout / 8 * f * Cout = 30 mV max
⇒ Cout = 3.75 uF

Cout > 4 * (Ioh - Iol) / f * ΔVout_shoot
Using Ioh = 1.7A, Iol = 1.3A, ΔVout_shoot = 250 mV:
Cout > 12.8 uF
Using Cout > 12.8 uF at 50 mΩ ESR: Choose Cout = 2 x 10 uF + 1 x 100 nF = 20.1 uF

Sheet: /1_Power/
File: 1_Power.kicad_sch
Title: Polaris/Power
Size: A4 Date:
KiCad E.D.A. 10.0.3
Rev:
Id: 2/11

[2] USB

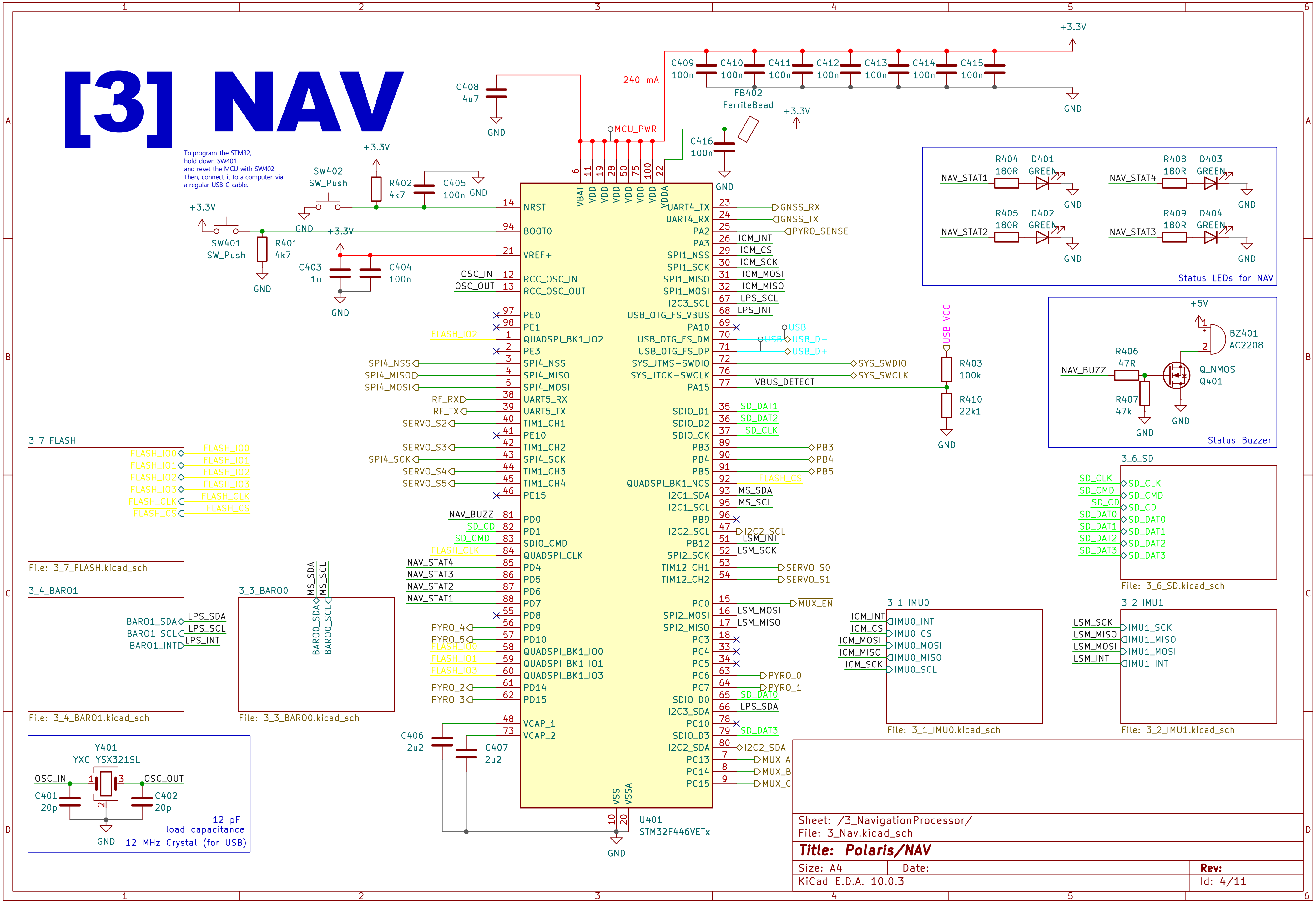


Sheet: /2_USB/
File: 2_USB.kicad_sch

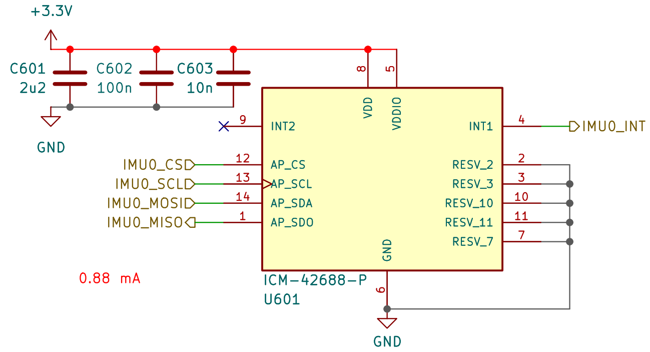
Title: Polaris/USB

Size: A4 | Date: | **Rev:**

KiCad E.D.A. 10.0.3 | Id: 3/11



[3] NAV
To program the STM32, hold down SW401 and reset the MCU with SW402. Then, connect it to a computer via a regular USB-C cable.
+3.3V
SW402
SW_Push
R402
4k7
C405
100n
GND
SW401
SW_Push
R401
4k7
C403
1u
C404
100n
C408
4u7
240 mA
MCU_PWR
C409 100n
C410 100n
C411 100n
C412 100n
C413 100n
C414 100n
C415 100n
FB402
FerriteBead
C416
100n
NRST
BOOT0
VREF+
OSC_IN
OSC_OUT
RCC_OSC_IN
RCC_OSC_OUT
VBAT
VDD
VDDA
PE0
PE1
FLASH_IO2
QUADSPI_BK1_IO2
PE3
SPI4_NSS
SPI4_MISO
SPI4_MOSI
RF_RX
RF_TX
UART5_RX
UART5_TX
SERVO_S2
TIM1_CH1
PE10
SERVO_S3
TIM1_CH2
SPI4_SCK
SERVO_S4
TIM1_CH3
SERVO_S5
TIM1_CH4
PE15
NAV_BUZZ
PD0
SD_CD
PD1
SD_CMD
SDIO_CMD
FLASH_CLK
QUADSPI_CLK
NAV_STAT4
PD4
NAV_STAT3
PD5
NAV_STAT2
PD6
NAV_STAT1
PD7
PD8
PYRO_4
PD9
PYRO_5
PD10
FLASH_IO0
QUADSPI_BK1_IO0
FLASH_IO1
QUADSPI_BK1_IO1
FLASH_IO3
QUADSPI_BK1_IO3
PYRO_2
PD14
PYRO_3
PD15
C406
2u2
C407
2u2
VCAP_1
VCAP_2
VSS
VSSA
U401
STM32F446VETx
UART4_TX
GNSS_RX
UART4_RX
GNSS_TX
PA2
PYRO_SENSE
PA3
ICM_INT
SPI1_NSS
ICM_CS
SPI1_SCK
ICM_SCK
SPI1_MISO
ICM_MOSI
SPI1_MOSI
ICM_MISO
I2C3_SCL
LPS_SCL
USB_OTG_FS_VBUS
LPS_INT
PA10
USB
USB_OTG_FS_DM
USB_D-
USB_OTG_FS_DP
USB_D+
SYS_JTMS-SWDIO
SYS_SWDIO
SYS_JTCK-SWCLK
SYS_SWCLK
PA15
VBUS_DETECT
USB_VCC
R403
100k
R410
22k1
SDIO_D1
SD_DAT1
SDIO_D2
SD_DAT2
SDIO_CK
SD_CLK
PB3
PB4
PB5
QUADSPI_BK1_NCS
FLASH_CS
I2C1_SDA
MS_SDA
I2C1_SCL
MS_SCL
PB9
I2C2_SCL
PB12
LSM_INT
SPI2_SCK
LSM_SCK
TIM12_CH1
SERVO_S0
TIM12_CH2
SERVO_S1
PC0
MUX_EN
SPI2_MOSI
LSM_MOSI
SPI2_MISO
LSM_MISO
PC3
PC4
PC5
PC6
PYRO_0
PC7
PYRO_1
SDIO_D0
SD_DAT0
I2C3_SDA
LPS_SDA
PC10
SDIO_D3
SD_DAT3
I2C2_SDA
PC13
MUX_A
PC14
MUX_B
PC15
MUX_C
R404 180R D401 GREEN NAV_STAT1
R408 180R D403 GREEN NAV_STAT4
R405 180R D402 GREEN NAV_STAT2
R409 180R D404 GREEN NAV_STAT3
Status LEDs for NAV
+5V
BZ401
AC2208
R406
47R
Q_NMOS
Q401
R407
47k
Status Buzzer
3_7_FLASH
FLASH_IO0
FLASH_IO1
FLASH_IO2
FLASH_IO3
FLASH_CLK
FLASH_CS
File: 3_7_FLASH.kicad_sch
3_4_BARO1
BARO1_SDA
BARO1_SCL
BARO1_INT
LPS_SDA
LPS_SCL
LPS_INT
File: 3_4_BARO1.kicad_sch
3_3_BARO0
BARO0_SDA
BARO0_SCL
File: 3_3_BARO0.kicad_sch
3_6_SD
SD_CLK
SD_CMD
SD_CD
SD_DAT0
SD_DAT1
SD_DAT2
SD_DAT3
File: 3_6_SD.kicad_sch
3_1_IMU0
IMU0_INT
IMU0_CS
IMU0_MOSI
IMU0_MISO
IMU0_SCL
File: 3_1_IMU0.kicad_sch
3_2_IMU1
IMU1_SCK
IMU1_MISO
IMU1_MOSI
IMU1_INT
File: 3_2_IMU1.kicad_sch
Y401
YXC YSX321SL
C401
20p
C402
20p
12 pF
load capacitance
12 MHz Crystal (for USB)
Sheet: /3_NavigationProcessor/
File: 3_Nav.kicad_sch
Title: Polaris/NAV
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.3
Id: 4/11



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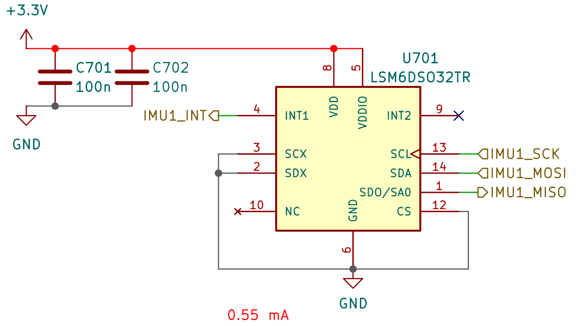
File: 3_1_IMU0.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 6/11

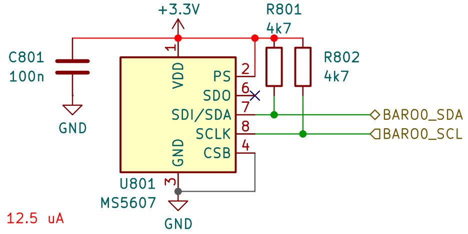


Sheet: /3_NavigationProcessor/3_2_IMU1/
File: 3_2_IMU1.kicad_sch

Title:

Size: A4 | Date: | **Rev:**

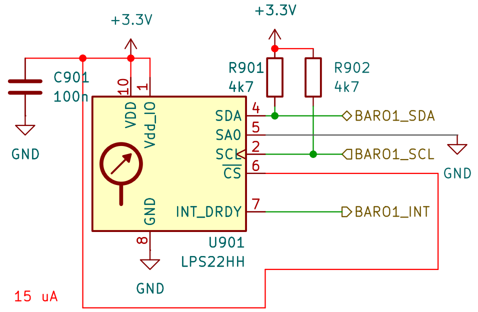
KiCad E.D.A. 10.0.3 | Id: 7/11



Sheet: /3_NavigationProcessor/3_3_BARO0/
File: 3_3_BARO0.kicad_sch

Title:

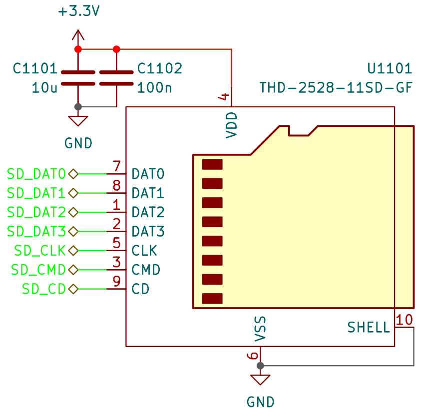
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 8/11



Sheet: /3_NavigationProcessor/3_4_BARO1/
File: 3_4_BARO1.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 9/11

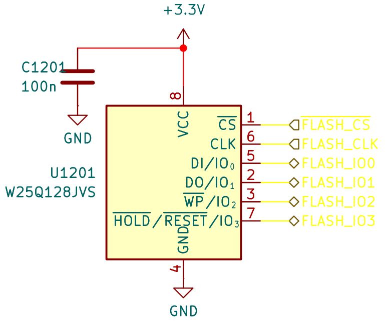


Sheet: /3_NavigationProcessor/3_6_SD/
File: 3_6_SD.kicad_sch

Title:

Size: A4 | Date: | **Rev:**

KiCad E.D.A. 10.0.3 | Id: 11/11



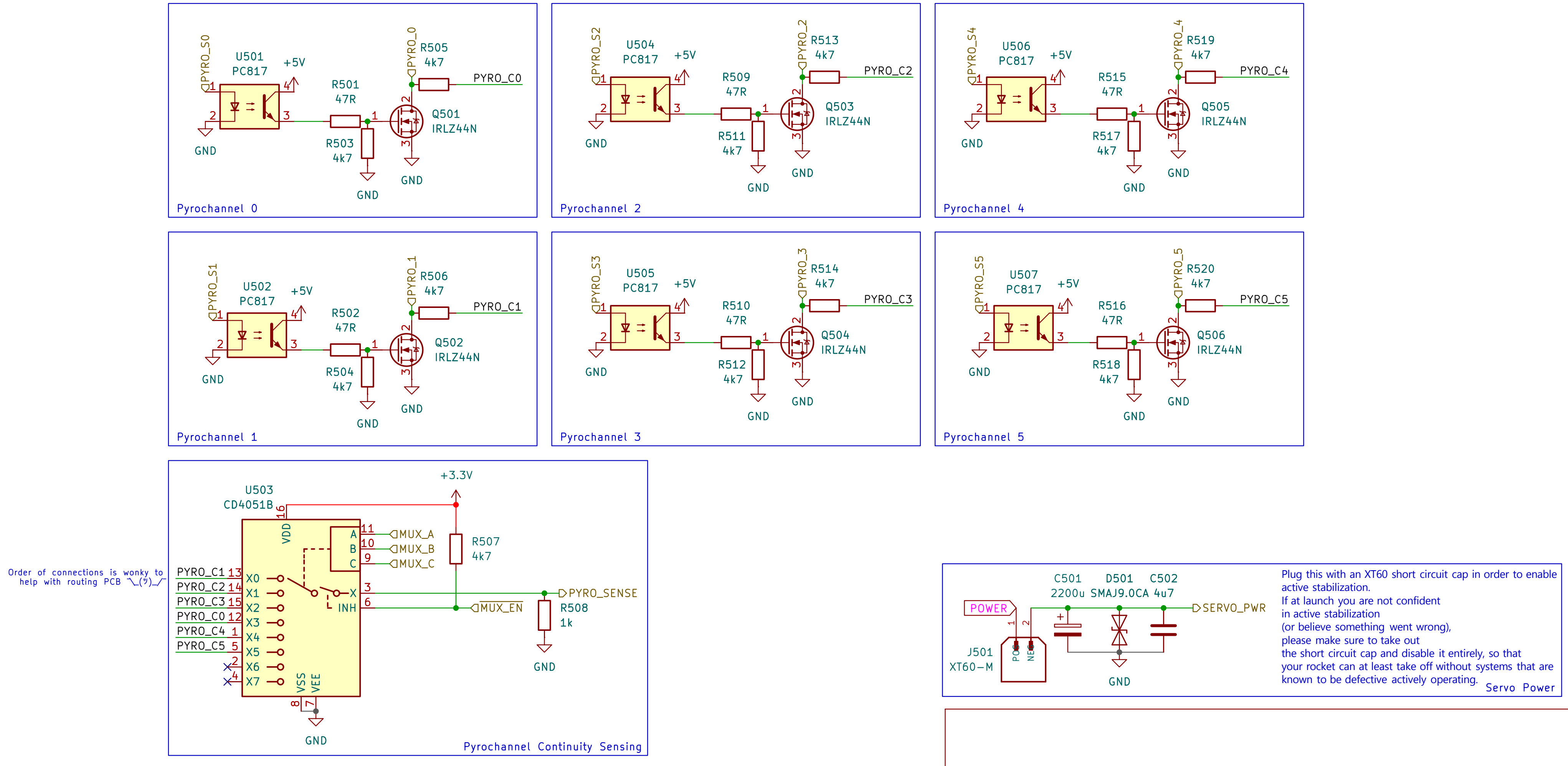
Sheet: /3_NavigationProcessor/3_7_FLASH/
File: 3_7_FLASH.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 12/11

[5] PYRO

20 mA total



Sheet: /4_Pyrochannels/
File: 4_Pyro.kicad_sch

Title: Polaris/Pyro

Size: A4 Date: Rev:
KiCad E.D.A. 10.0.3 Id: 5/11